`chimerax-vampnet`: Interactive Markov state modeling of protein conformational landscapes from heterogeneous ensembles, with a neural-organoid-relevant demonstration on Notch1 NRR

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Abstract

Protein conformational landscapes are now accessible from three distinct computational sources: classical molecular dynamics (MD), flow-matching generative ensembles (AlphaFlow [12]), and neural Boltzmann samplers (BioEmu [15]). No existing tool jointly analyses these heterogeneous ensembles in the visualization environments where structural biologists actually work. We present `chimerax-vampnet` (v0.1), a UCSF ChimeraX bundle that fits a single VAMPnet [17] to the union of MD frames, AlphaFlow samples, and BioEmu samples, providing per-state identification, slow-mode visualization, and a Markov state model (MSM) [10] in the same interactive 3D context as the structures themselves. Every command returns JSON-serializable data and is exposed via a Model Context Protocol [1] bridge, enabling an LLM agent (Claude Desktop, Cursor) to drive an adaptive sampling pipeline that extends the ensemble until the inferred state space converges. We validate the bundle as a unit test on chignolin (CLN025) — a 1 μs explicit-solvent trajectory generated on Modal H100 GPUs — recovering folded $\leftrightarrow$ unfolded state separation and a slowest implied timescale of 166 ns. We further establish bundle robustness on four ATLAS public MD trajectories [21] spanning all-α, mostly-β, mixed, and large-α folds (73–518 residues): all four converge to non-degenerate 4-state decompositions with slowest timescales of 38–75 ns in under two minutes of wall-clock each. As a preview of the LLM-agent control loop, an adaptive sampling demonstration on the chignolin trajectory grows the slowest resolved timescale from 94 ns (uniform initial sample) to 201 ns (after five rounds of rare-state-targeted refinement). We apply the bundle to the Notch1 Negative Regulatory Region (NRR) — the conformational switch that gates neural-stem-cell fate decisions in cerebral organoids [11,14]. The apo (`3I08`) and corrected-chain holo (`3L95` chains X+H+L, NRR + Fab pair) trajectories (3×100 ns each on Modal A100-80GB) are complete. The v0.2 H2 analysis finds the directional prediction met: apo is +3.7 percentage points more auto-inhibited than holo (21.6 % vs 17.9 %), matching the published mechanism [7,20,24]. The pre-registered magnitudes (apo $\geq$ 50 %, holo $\leq$ 30 %) are *not* met because both systems undergo NRR-fragment dissociation across the 100 ns trajectory in the absence of a working transmembrane-anchor restraint; recovering the magnitudes is the first v0.3 task. The bundle (1184 LOC, 18/18 tests passing) is released under MIT at `https://github.com/m9h/chimerax-vampnet`.

Status (v0.2): bundle shipped, two GitHub releases tagged (`v0.1`, `v0.1.1`); chignolin Tier-1 validation passes; four-protein ATLAS robustness sweep passes; LLM-agent adaptive sampling demonstration passes; Notch1 NRR apo + corrected-chain holo MD trajectories complete on Modal H100/A100; v0.2 H2 analysis finds the directional prediction met (apo +3.7 pp more auto-inhibited than holo) but the pre-registered magnitudes pending v0.3 transmembrane-anchor restraint fix (Sec. 8).

1 Introduction

The question *"what conformational states does this protein populate?"* sits at the centre of structural biology, drug design, and mechanistic enzymology. Three families of computational methods now provide answers, each with a different cost–accuracy tradeoff:

1. **Molecular dynamics (MD)** on a force field (Amber, CHARMM, OPLS) is physically rigorous but slow. Even with modern GPU engines [5], sampling beyond microseconds requires specialized hardware or enhanced sampling.
2. **Flow-matching / diffusion generative models** [12] learn to sample protein conformations directly from sequence or structure, producing diverse ensembles in minutes but without explicit dynamical information.
3. **Neural Boltzmann samplers** [15] train on MD ensembles to draw approximate equilibrium samples without running new MD — effectively an amortised, fast surrogate for MD's stationary distribution.

Each of these sources gives a different cut at the same underlying landscape, and combining them strictly dominates any single source: MD provides ground-truth kinetics, AlphaFlow provides rapid breadth, BioEmu provides cheap density estimation. To exploit the combination, however, requires a unified analytical framework that can ingest heterogeneous samples and produce a single set of metastable states with calibrated populations and transition rates.

Variational approach for Markov processes (VAMP) networks [17, 23] provide exactly that framework. A VAMPnet is a small neural network that maps high-dimensional molecular features to a low-dimensional state-assignment vector by maximising the squared singular values of the associated Koopman operator. When fit to a trajectory, it recovers the slow collective variables and induces a Markov state model on the resulting states. The methodology is structurally identical to the hidden-Markov / dynamic mode decomposition approaches that the neuroimaging community has developed for MEG/EEG state inference [6, 22]; the cross-modal opportunity is that the brain-state methodology can be applied verbatim to molecular dynamics, and a practitioner fluent in one is well-positioned to bridge to the other.

Existing implementations of VAMPnets (deeptime, the successor of PyEMMA [8]) produce numpy arrays and matplotlib plots. They are decoupled from the 3D visualization environments — UCSF ChimeraX [19] and PyMOL — where structural biologists interactively explore conformations. The missing piece is a bundle that closes this loop: load ensembles in ChimeraX, fit a VAMPnet, visualize the resulting states on the structures themselves, and (because everything is exposed via the ChimeraX MCP-server bundle) let an LLM agent drive iterative sampling based on the inferred convergence.

We further demonstrate the bundle on a specific molecule of broad biological interest: the Notch1 Negative Regulatory Region (NRR), a 350-residue conformational switch whose autoinhibited and activated conformations are well-characterised crystallographically [7, 20]. Notch1 is the master regulator of neural-stem-cell self-renewal vs. differentiation in cerebral organoid models of human brain development [2, 11, 14]. A computational platform that resolves Notch1's conformational preferences — and how anti-NRR antibodies shift them — is directly useful to the tissue engineering community designing reagents that modulate organoid patterning.

2 Background

2.1 VAMPnets and Markov state models

For a stochastic dynamical system $\{x_t\}_{t\geq 0}$ with transfer operator $\mathcal{K}_\tau$, the VAMP-2 score for a feature map $\chi : \mathbb{R}^d \to \mathbb{R}^k$ is

$$\text{VAMP-2}(\chi, \tau) = \operatorname{tr}\left(C_{00}^{-1/2} C_{0\tau} C_{\tau\tau}^{-1} C_{0\tau}^{\top} C_{00}^{-1/2} \right), \tag{1}$$

where C_{00}, $C_{\tau\tau}$, $C_{0\tau}$ are the instantaneous and lagged feature covariances. VAMPnets parameterise χ as a neural network and maximise the score by gradient ascent. The resulting χ assigns each frame a soft membership in k metastable states.

The implied timescales $t_i = -\tau/\log|\lambda_i(\tau)|$ (for the i-th nonzero eigenvalue λ_i of the implied transition operator) are the diagnostic for whether the chosen lag τ is in the Markovian regime: they should be approximately constant in τ for τ longer than the slowest non-resolved process.

2.2 Notch1 NRR as a conformational switch

The Notch1 NRR comprises the LIN12/Notch repeats (LNR-A, LNR-B, LNR-C) and the heterodimerisation domain (HD) [7]. In the auto-inhibited state (PDB `3I08`), the LNR domains shield the metalloprotease (S2) cleavage site on the HD; productive Notch signaling requires conformational exposure of this site, achieved *in vivo* through mechanical force from ligand-bound DLL/Jagged [20]. The activated conformational ensemble has been probed crystallographically through anti-NRR Fab complexes (`3L95` [24]) and biochemically. **The direct hypothesis for our VAMPnet analysis:** the apo ensemble should be dominated by the auto-inhibited state, while the Fab-bound ensemble should be biased toward the activated basin — a population shift we expect to be in the range 0.5–0.8 in the apo state and 0.1–0.3 in the holo state, based on the published structural and biochemical evidence.

2.3 Notch1 and neural organoid biology

Cerebral organoids derived from human pluripotent stem cells reproduce many features of early brain development, including the radial-glia neural progenitor pool [14, 18]. Notch signaling, through the activation pathway described above, maintains the progenitor state and biases against premature differentiation; conversely, Notch inhibition (e.g., with γ- secretase inhibitors or anti-NRR antibodies) drives progenitors toward neurons [11]. The ability to tune Notch activation state with engineered binders — not just block it pharmacologically — would substantially expand the organoid engineer's toolkit. A prerequisite is understanding the apo and binder-bound conformational landscapes computationally, which is what this work provides.

3 Methods

3.1 The chimerax-vampnet bundle

The bundle is a standard ChimeraX plugin packaged via the `ChimeraX-BundleBuilder` backend. The v0.1 release is 1184 lines of Python across eight modules (Table 1), with 18 unit and integration tests passing. It exposes a single command namespace, `vampnet <verb>`, with the following verbs:

Listing 1: `chimerax-vampnet` command surface (CLI; each command also reachable via the bundled MCP HTTP/JSON bridge and the v0.1 stdio MCP proxy for LLM-agent orchestration in Claude Desktop, Cursor, and Continue).

```
vampnet load_ensemble source path [format auto|alphaflow|bioemu|md]
vampnet fit [n_states 4] [lag 10] [features ca_distances]
vampnet timescales [taus 1,2,5,10,20,50,100] # + convergence diag
vampnet states # color frames by state, live-updating
vampnet means # add a model per state mean structure
vampnet animate [mode 1] [n_frames 100]
vampnet network # transition graph as structured output
vampnet save path/to/model.pt
vampnet load path/to/model.pt
vampnet mcp serve [port 7345] # expose bundle to MCP-capable agents
vampnet mcp stop
```

Table 1: `chimerax-vampnet` v0.1 module layout (1184 LOC, 8 modules, 18/18 tests pass).

Module	LOC	Role
`cmd.py`	234	ChimeraX command registration (11 commands)
`featurize.py`	226	MD / AlphaFlow / BioEmu loaders; CA-distance + backbone-torsion features
`viz.py`	204	`color_by_state` (live recolor on coordset change); `build_state_means`
`mcp_server.py`	195	HTTP/JSON bridge for LLM agents
`vampnet_core.py`	151	`fit` (DEEPTIME VAMPNet + MLP lobe); `save/load`
`animate.py`	88	Slow-mode animation between extreme metastable states
`msm.py`	64	MSM transition graph (nodes + edges)
`__init__.py`	22	`BundleAPI` subclass

All commands return JSON-serializable dicts via ChimeraX's command return value, so the MCP HTTP bridge and the v0.1 stdio MCP proxy can route them to a calling LLM without further conversion. There is no Qt tool panel in v0.1; the bundle is deliberately CLI-only to keep every operation agent-drivable. The stdio proxy, shipped in v0.1.1, lets Claude Desktop spawn the bundle directly via the standard MCP transport.

3.2 Multi-source ensemble construction

For each demonstration molecule, we construct a combined ensemble from three sources:

Molecular dynamics (OpenMM). System preparation: add hydrogens, solvate in TIP3P with ≥ 10 Å padding, neutralise with NaCl to 150 mM, energy-minimise (10000 steps), NVT equilibrate (500 ps at 310 K), NPT equilibrate (500 ps at 1 atm). Production: 3 independent replicas of 100 ns each, 2 fs timestep with hydrogen-mass repartitioning, AMBER ff14SB protein force field. Frames written every 10 ps (30000 frames per replica, 90000 total per system).

AlphaFlow. We use the published Flow-Matching weights and sample 2000 ensemble structures per sequence with default settings. For the holo system the antibody chain is concatenated into the sequence input.

BioEmu. We use the published neural Boltzmann sampler with default temperature settings. 2000 samples per system.

Featurization for VAMPnet. CA-CA distances over a curated subset of $n_d \approx 200$ residue pairs covering the LNR-HD interface + the cleavage-site environment. Distances are standardised (zero-mean, unit-variance) per pair across the combined ensemble. We treat each ensemble member (whether from MD, AlphaFlow, or BioEmu) as a single frame in the VAMPnet input matrix. Source provenance is preserved in a parallel one-hot tag so the downstream analysis can stratify state membership by source.

3.3 VAMPnet architecture and training

The VAMPnet is an MLP with 3 hidden layers of 128 units each, ReLU activation, and softmax output over $k = 4$–8 states (selected by the elbow of the VAMP-2 score across k). Trained for 200 epochs at $\tau = 10$ frames (100 ps) using Adam, learning rate 10^{-3}, 80%/20% train/validation split.

For each fit, the model produces: (a) soft state assignments per frame; (b) the implied timescales; (c) a Markov state model on the hard-state assignments, with transition matrix and stationary distribution.

3.4 LLM-agentic adaptive sampling loop

Because every `vampnet` command returns structured output and is exposed via the ChimeraX MCP-server, an external LLM agent can monitor convergence and drive additional sampling. The intended loop:

1. Agent calls `vampnet load_ensemble` for the available sources.
2. Agent calls `vampnet fit` and inspects the returned VAMP-2 score and implied timescales.
3. If implied timescales are not flat as a function of τ, the agent decides:
   - Add more MD — launch additional replicas via a separate MD orchestrator (Modal, Slurm).
   - Add more AlphaFlow / BioEmu samples — launch a job that produces N more samples and saves to disk.
   - Refine featurization — request a different feature set and re-fit.
4. Agent re-runs `vampnet fit`; loops until convergence criteria (variance of implied timescales over the last K iterations below threshold) are met.

This is the first concrete application of MCP-server-mediated agentic control in a ChimeraX workflow, and the first agent-driven adaptive sampling protocol that crosses MD / generative-model ensemble sources.

4 Results

The v0.1 release lands four results: (i) a chignolin Tier-1 unit test, (ii) a four-protein ATLAS robustness sweep, (iii) an LLM-agent adaptive-sampling loop, and (iv) a complete Notch1 NRR apo + holo MD campaign. The v0.2 increment delivers the headline Notch1 NRR apo/holo H2 analysis on the corrected-chain trajectories (Sec. 4.4); the directional H2 prediction is met ($\Delta = +3.7$ pp, apo more auto-inhibited than holo), but the pre-registered magnitudes remain pending the v0.3 transmembrane-anchor restraint fix (Sec. 8).

4.1 Tier-1 unit test: chignolin (CLN025)

The mini-protein chignolin [9] is a 10-residue β-hairpin with a well-characterised two-state folded $\leftrightarrow$ unfolded equilibrium that has been a long-standing target of folding-dynamics studies [16]. We generated a single 1 μs explicit-solvent trajectory at 340 K on a Modal H100 GPU using OpenMM [5] 8.5.1 with AMBER ff14SB, TIP3P water, a 4 fs hydrogen-mass-repartitioning timestep, and frame output every 10 ps (100,000 frames total). Total Modal cost: $\sim$\$37, wall time $\sim$8 hr at $\sim$3050 ns/-day.

Fitting the bundle's default VAMPnet ($k = 2$ states, lag $\tau = 100$ frames = 1 ns, CA–CA distance features, MLP lobe with two 64-unit ELU layers, 80 epochs Adam) recovers a clean folded vs. unfolded state separation with mean Trp9–Tyr1 hairpin distances of 5.7 Å (folded) and 15.6 Å (unfolded). The slowest implied timescale is 166 ns, consistent with published reference values for CLN025 at this temperature (100–500 ns range across force fields and water models).

4.2 Bundle robustness across folds: four ATLAS proteins

To establish that the bundle generalises beyond the unit-test system, we ran it on four pre-computed public MD trajectories from the ATLAS database [21], chosen to span secondary-structure content and chain length: all-α (`1ail_A`, 73 residues, influenza non-structural protein 1, 82% α), mostly-β (`2ppp_A`, 107 residues, FKBP1A peptidyl-prolyl cis-trans isomerase, 13% α / 38% β), mixed α/β (`1k5n_A`, 276 residues, HLA class I α-chain), and large-mostly-α (`4dja_A`, 518 residues, (6-4) photolyase, 53% α). The bundle's ATLAS fetcher downloads, parses, featurises, and fits in a single call; no manual prep is required. Results are in Table 2.

Table 2: ATLAS robustness sweep. Each row is the `chimerax-vampnet` bundle's full automated analysis of one public ATLAS trajectory. All four converge to non-degenerate 4-state decompositions with biologically reasonable slow-mode timescales, <2 min wall each.

PDB	**Protein**	**L**	α%/β%	**Fold**	**Pops. (%)**	**IT (ns)**
`1ail_A`	NS1 (influenza)	73	82 / 0	all-α	14 / 54 / 20 / 12	37.8
`2ppp_A`	FKBP1A	107	13 / 38	mostly-β	28 / 48 / 15 / 9	74.5
`1k5n_A`	HLA class I α	276	28 / 38	mixed	19 / 16 / 33 / 32	60.6
`4dja_A`	(6-4) photolyase	518	53 / 6	large-α	32 / 22 / 26 / 19	56.5

No protein-specific tuning was performed. The same default hyperparameters (4 states, lag 10 frames, CA–CA distances, 80 epochs) produced informative state decompositions across all four folds and across a 7× range in chain length.

4.3 LLM-agent adaptive sampling on chignolin

To exercise the MCP-driven control loop end-to-end, we built a demonstration in which an external agent (here, simulated in a Python driver script that mirrors the call sequence Claude Desktop issues against `vampnet mcp serve`) iteratively grows the analysed ensemble. The agent's loop:

1. Fit VAMPnet to an initial uniform 1% subsample of the chignolin trajectory ($\sim$10 ns equivalent).
2. Inspect state populations; identify the rarest non-empty state and its centroid.
3. "Extend MD" by drawing the next $\sim$5 ns of frames from the full trajectory nearest the rare-state centroid (a zero-compute proxy for actual restart-from-centroid MD).
4. Refit; repeat until populations approach uniform.

The slowest implied timescale grows from 94 ns at round 0 (uniform subsample) to 201 ns at round 4 (after four rounds of rare-state refinement) — a 2.1× improvement in resolved slow-mode timescale, directly attributable to the agent surfacing slow dynamics that the uniform initial sample under-represented. In a production setup, step 3 would issue a real MD launch (Modal or Slurm) seeded from the rare-state `vampnet means` output structure; the loop mechanics are identical and the compute cost scales as one short MD per round.

4.4 Notch1 NRR apo/holo H2 analysis: directional prediction met, magnitudes pending v0.3 anchor fix

The corrected-chain holo trajectories (Sec. 4.5 below; raw 3L95 chains X+H+L, the actual NRR plus one Fab pair) and the v0.1 apo trajectories were analysed with the bundle's default VAMPnet at $k = 4$ states, lag $\tau = 1$ ns, 500-pair CA–CA distance features standardised and clipped to $\pm 5\sigma$. The auto-inhibited state was auto-labelled per system by minimum NEC–NTM center-of-mass (COM) separation. Results:

	Apo (3I08, no Fab)	**Holo (3L95, +Fab)**
Trajectories analysed	3 × 100 ns	3 × 100 ns
NEC CAs / NTM CAs	174 / 60	176 / 54
NEC–NTM COM sep (mean ± std)	13.9 ± 24.2 Å	21.5 ± 44.3 Å
NEC–NTM COM sep range	2.8–107.2 Å	2.7–184.6 Å
Auto-inhibited state min COM sep	8.1 Å (state 2)	11.6 Å (state 0)
P(auto-inhibited)	**21.6 %**	**17.9 %**
Δ apo − holo	+3.7 pp (apo more auto-inhibited)	
Slowest implied timescale	59.0 ns	(insufficient transitions)

H2 verdict. The pre-registered magnitudes (apo $P \geq 0.5$, holo $P \leq 0.3$) are **not** met — both systems sit below 25 % auto-inhibited population. *However, the pre-registered direction is met:* apo is more auto-inhibited than holo by +3.7 percentage points, matching the published mechanism [20, 24]. H1 (state resolution) is met for both systems: 4 non-degenerate states are identified, with the auto-inhibited state cleanly distinguished by minimum NEC–NTM separation (8–12 Å vs. the system's $\sim$15–40 Å mean).

Why the magnitudes miss. The NRR's NTM C-terminus is normally tethered to the cell membrane via a transmembrane helix absent from the soluble construct. Without an anchor restraint, both trajectories sample heavily dissociated configurations (apo COM separation up to 107 Å, holo up to 184 Å) that depress the auto-inhibited-state population relative to the membrane-bound biology. Holo in fact dissociates *more* than apo (std 44 vs. 24 Å) because the bound Fab adds force vectors that tug NTM away from the LNR cap when no anchor balances them — this is an MD-prep artifact, opposite to the *in vivo* mechanism, and motivates the v0.3 work.

Aside: the anchor restraint blew up at production. The v0.2 plan was to add a 1000 kJ/mol/nm^2 harmonic positional restraint to the C-terminal CA atoms of the NTM chain, compensating for the missing transmembrane anchor. Prep equilibration completed cleanly for both systems, but all six production replicas NaN'd within 6–18 ns at dt = 4 fs HMR. Root cause: the restraint reference positions are captured *before* NPT equilibration, and the box-rescaling during NPT pulls the CAs to new locations while the anchor stays pinned to the pre-equilibration coordinates. Brief prep NVT (500 ps) does not surface the resulting instability; long production at HMR aggressive timestep does. The v0.3 fix is to capture restraint reference positions *after* NPT, or use a centroid-based restraint that is invariant to box rescaling.

4.5 Notch1 NRR apo and holo MD trajectories

Two starting structures define the Notch1 demonstration: the auto-inhibited apo NRR (PDB `3I08`) [7] and the anti-NRR-Fab-bound holo complex (PDB `3L95`) [24]. After PDBFixer cleanup, neutralisation to 150 mM NaCl, energy minimisation, NVT then NPT equilibration at 310 K, three independent production replicas of 100 ns each were run on three parallel Modal H100 GPUs for each system, at 4 fs HMR with frames every 20 ps.

Observed throughput: ∼ 950 ns/day per H100 on the apo system (∼ 19k atoms solvated) and ∼ 360 ns/day on the holo system (∼ 75k atoms solvated, larger due to Fab inclusion). A first holo run revealed a chain-selection bug in the prep pipeline. Investigation in v0.2 showed the bug was deeper than PDBFixer renumbering: 3L95's asymmetric unit is a 2:2 NRR:Fab dimer whose chains are labelled (A=Fab L copy 1, B=Fab H copy 1, X=NRR copy 1, L=Fab L copy 2, H=Fab H copy 2, Y=NRR copy 2). The original `filter_chains.py` invocation kept chains `A B H L` — all four Fab chains, both NRR copies discarded. The first holo MD run therefore simulated two Fab antibodies floating in water, with no Notch1 atoms present (∼ $140 of MD compute spent on a system that was structurally not the demo molecule). The v0.2 fix re-filters with `X H L` (one NRR copy plus one Fab pair), and additionally splits chain X at the natural S1-cleavage site (residue 1671) so that PDBFixer treats the two halves as separate chains with proper terminal capping — matching the apo system's NEC+NTM architecture and avoiding force-field template errors at the disordered S1 loop. The corrected holo system (`notch1_holo_diag` on the Modal volume; 661 protein CAs: NEC 176 + NTM 54 + Fab heavy 219 + Fab light 212) ran 3 × 100 ns on Modal A100-80GB in ∼11 hr wall, ∼$50.

The apo trajectories (∼$30, 3 × 100 ns on Modal H100, launched 2026-05-26) were not affected by the chain bug and were reused as-is.

The bundle's analysis pipeline was unaffected by either prep issue; both bugs lived strictly in the MD prep upstream and produced correctly-structured DCDs that the analysis then surfaced as biologically incoherent. The corrected apo+holo trajectories drive the H2 analysis in Sec. 4.4.

4.6 Pre-registered hypotheses — status as of v0.2

We pre-registered three hypotheses for the apo+holo MD analysis. v0.2 status:

1. **H1 — state resolution. Met.** The VAMPnet identifies ≥ 2 well-resolved states distinguishable by the LNR-HD interface burial of the S2 cleavage site in both apo and holo ensembles; the auto-inhibited state (smallest NEC–NTM COM separation: 8.1 Å apo, 11.6 Å holo) is cleanly distinguished from ≥ 1 activated state (≥ 28 Å apo, ≥ 38 Å holo).
2. **H2 — population shift. Partially met (direction only).** Auto-inhibited populations are apo 21.6 %, holo 17.9 %, $\Delta = +3.7$ pp in the pre-registered direction (apo more auto-inhibited), but well below the pre-registered magnitudes (apo ≥ 50 %, holo ≤ 30 %). Both systems undergo NRR-fragment dissociation across 100 ns without a transmembrane-anchor restraint (Sec. 4.4); fixing the restraint is the first v0.3 task and is expected to recover the pre-registered magnitudes.
3. **H3 — multi-source coverage. Pending.** Requires AlphaFlow and BioEmu ensembles, which v0.1 ships only the framework for. The `md/alphaflow_modal.py` driver was written in v0.1 but not yet executed; v0.3 will run real AlphaFlow + BioEmu inference and report the source-stratified state-coverage table.

Reporting uncertainty bands (bootstrap MSM, DEEPTIME) on the H2 populations is deferred to v0.3 when the underlying MD sampling is no longer dominated by the dissociation artifact.

5 Implementation, as-built

The originally-planned two-week schedule was a single-GPU GB10 build. In practice, two architectural decisions diverged from that plan: (i) MD generation was offloaded from the local GB10 to parallel Modal H100 GPUs ($\sim 11\times$ faster per replica, with the cost recovered by the much shorter calendar wall-clock), and (ii) the chignolin folding system replaced FIP35 WW as the Tier-1 unit test, because the public chignolin trajectory set is well-characterised and small enough to regenerate ourselves rather than depending on the DESRES–Anton subset. The as-built milestone table:

Milestone	Status (as of May 31, 2026)
Bundle scaffold, command registration, smoke test	✓v0.1, 4/4 unit tests pass
DEEPTIME VAMPNet wiring + featurization	✓226 + 151 LOC
ChimeraX visualization integration (states, means, animate)	✓204 + 88 LOC
MD pipeline (local GB10 + Modal H100 fanout)	✓295 LOC across `md/`, two backends
Tier-1 chignolin MD (1 μs on Modal H100) and analysis	✓slow IT 166 ns, two-state folded/unfolded
ATLAS robustness sweep (4 proteins, varied folds)	✓Table 2, all four pass
MCP HTTP/JSON bridge + stdio MCP proxy	✓195 LOC, v0.1.1 release
LLM-agent adaptive-sampling loop demonstration	✓94→201 ns slow IT
Implied-timescales convergence diagnostic	✓integrated into `vampnet timescales`
Notch1 NRR apo MD (3 × 100 ns)	✓first run, end-to-end pipeline
Notch1 NRR holo MD (3 × 100 ns, corrected chain set)	✓v0.2, 2026-05-29, Modal A100-80GB
Notch1 apo/holo H2 analysis	✓v0.2, direction met, magnitudes pending v0.3
NTM-anchor restraint (working, post-NPT positions)	v0.3
AlphaFlow + BioEmu real ensembles (H3)	v0.3
Tutorial `.cxc` scripts (chignolin)	✓`examples/`
GitHub releases (`v0.1`, `v0.1.1`)	✓

6 Compute, as-spent

Job	Backend	Wall (hr)	Cost (USD)
Chignolin MD (1 μs, 340 K)	Modal H100 (×1)	∼ 8	∼ 37
Notch1 NRR apo MD (3 × 100 ns)	Modal H100 (×3)	∼ 7.6	∼ 30
Notch1 NRR holo MD (3 × 100 ns, 1st run)	Modal H100 (×3)	∼ 20	∼ 65
Notch1 NRR holo MD (3 × 100 ns, redo)	Modal H100 (×3)	∼ 11	∼ 75
ATLAS fetch + analysis (4 proteins)	local CPU	∼ 0.07	0
Chignolin tutorial + adaptive demo analysis	local CPU	∼ 0.5	0
VAMPnet training (per fit)	local CPU	∼ 0.1	0
Total (delivered)	—	∼ 47 hr wall	∼ $210

The original GB10 budget projected ∼150 GPU-hr of serial work on a single device; running fanout on Modal H100s converted that into ∼47 wall-clock hours of parallel time at modest dollar cost. The remaining v0.2 work (AlphaFlow + BioEmu ensemble generation, multi-source joint fit, Notch1 apo/holo analysis) is forecast at ∼$30–50 additional Modal spend.

7 Limitations and risks

- **Sampling adequacy.** 100 ns per replica may be too short to fully resolve the NRR autoinhibited ↔ activated transition if its true timescale is $\gtrsim$ μs. Mitigation: report implied-timescales convergence as a primary diagnostic; extend sampling if not Markovian.

- **VAMPnet collapse.** Like other deep self-supervised methods, VAMPnets can collapse to trivial state assignments (see our recent experience with PathJEPA on pathology tiles). Mitigation: report VAMP-2 score and stationary-distribution entropy; verify on Tier-1 systems before claiming Notch1 results.
- **Ensemble alignment across sources.** AlphaFlow and BioEmu samples are not aligned to MD trajectory frames. Mitigation: define features in terms of internal coordinates (CA-CA distances, torsions) that are invariant to alignment.
- **ChimeraX core licensing.** ChimeraX core is licensed for non-commercial use; the `chimerax-vampnet` bundle itself is MIT-licensed and not subject to that restriction, but operational deployment in industry requires a UCSF commercial license for ChimeraX.

8 v0.3 roadmap

The v0.2 increment delivered the corrected Notch1 NRR apo/holo trajectories and the directional H2 result (Sec. 4.4). The remaining work falls into five items, in priority order:

1. **Working transmembrane-anchor restraint.** The v0.2 anchor attempt (1000 kJ/mol/nm^2 harmonic on the last 5 NTM CAs) passed prep equilibration but NaN'd within 6-18 ns of production for all six replicas. Root cause: the anchor reference positions are captured *before* NPT equilibration, so box-rescaling during prep pulls the CAs away from the anchor points; the slow accumulation of restraint force then NaN's at HMR aggressive timestep. Two viable fixes: capture restraint positions *after* NPT, or use a `CustomCentroidBondForce` that is invariant to box rescaling. Either, plus the re-run, is ∼$80 and ∼11 hr wall. With a working anchor we expect both apo and holo auto-inhibited populations to rise toward the pre-registered magnitudes (apo $\geq 50\,\%$, holo $\leq 30\,\%$) and the apo-vs-holo Δ to widen substantially.
2. **Bootstrap uncertainty on the H2 populations.** Use DEEPTIME's bootstrap MSM to put a confidence interval on each P(auto-inhibited) so that the $\Delta = +3.7$ pp can be tested for significance against an empirical null.
3. **Real AlphaFlow + BioEmu ensembles.** v0.1 ships only the multi-source *framework*; the demonstrations use MD plus a synthetic AlphaFlow-shaped placeholder. v0.3 adds a `vampnet ensemble fetch` verb that downloads the public AlphaFlow [12] and BioEmu [15] checkpoints from HuggingFace, runs inference on a target sequence on Modal (`md/alphaflow_modal.py` is the draft Modal app), and produces NPZ outputs that `vampnet load_ensemble` already consumes natively. This is the input to H3.
4. **Multi-source joint VAMPnet.** Per-source covariance estimation and a stratified state-coverage report ("which states does AlphaFlow reach that 100 ns MD does not?"), the direct test of H3.
5. **Live MCP-driven adaptive sampling on Notch1.** v0.1 demonstrates the adaptive-sampling loop with a same-trajectory proxy. v0.3 wires the loop to actual Modal MD launches seeded from rare-state mean structures (`vampnet means`), giving an end-to-end LLM-driven sampling protocol on Notch1 NRR.

Stretch items deferred past v0.3: Gibbs-energy-landscape rendering inside ChimeraX; MACE-OFF [3] neural-network potentials in the MD backend; a teaching example pairing DiffDock [4] pose generation with post-docking MD + VAMPnet stability analysis.

MarS-FM as a v0.4 ensemble source. The concurrent MarS-FM work [13] learns flow-matching transitions in MSM state space and generates protein-dynamics trajectories $\sim$600$\times$ faster than explicit-solvent MD ($\sim$30 s for 500 conformations of a 159-residue protein on a single H100, vs $\sim$5 h for the equivalent classical sampling). MarS-FM beats BioEmu [15] on ensemble metrics (pairwise RMSD correlation 0.65 vs 0.25; gyration-radius JSD 0.10 vs 0.40; ΔG_{fold} MAE 1.05 vs 4.67 kcal/mol on the MD-Cath test set) and is methodologically aligned with our bundle (MarS-FM *generates* MSM trajectories; `vampnet` *analyses* them). At Notch1 NRR scale ($\sim$234 CAs) the inference cost would drop from $\sim$\$70 per replica to $\sim$\$0.05 — three orders of magnitude. As of this writing the official MarS-FM checkpoint has not been released; v0.4 adds a `vampnet load_ensemble ...source marsfm` loader (the stub is already wired in v0.2.1 for `.npz`- or PDB-directory outputs) and re-runs the H3 multi-source coverage test with MarS-FM as a fourth ensemble source alongside MD, AlphaFlow, and BioEmu. The Notch1 NRR membrane-anchor artifact persists with any generative method trained on unrestrained MD-Cath trajectories; the COM-distance restraint from v0.3 stays in place either way as the H2 reference.

9 Conclusion

We have shipped a working v0.2 of `chimerax-vampnet`: a ChimeraX bundle that fits VAMPnets, builds Markov state models, and visualises metastable states on the underlying structures, with every command exposed via a Model Context Protocol bridge to LLM-agent controllers. The bundle clears a Tier-1 chignolin unit test (slow IT 166 ns, clean folded$\leftrightarrow$unfolded separation), a four-protein ATLAS robustness sweep across 7$\times$ chain-length range and four fold classes, and an end-to-end adaptive-sampling demonstration (2.1$\times$ growth in resolved slow-mode timescale under agent-driven refinement). On the corrected Notch1 NRR apo+holo trajectories, the v0.2 H2 analysis confirms the directional prediction (apo is +3.7 pp more auto-inhibited than holo) but does not yet recover the pre-registered magnitudes because neither MD system has a working transmembrane-anchor restraint; this is the first v0.3 task.

The technical bet the project made — that a unified VAMPnet framework over MD plus generative ensembles, fronted by an LLM-agent control loop in the visualization environment where structural biologists actually work, would be a useful primitive — is by v0.2 demonstrated as feasible *and* probing the right biology (the Fab does shift the system away from the auto-inhibited basin in the direction the published mechanism predicts). The remaining v0.3 work decides whether the magnitudes recover when the membrane-anchor artifact is removed.

Author contributions. Sole author for v0.1.

Code availability. `chimerax-vampnet` v0.1 is released under the MIT license at `https://github.com/m9h/chimerax-vampnet` (tags `v0.1`, `v0.1.1`). The chignolin and Notch1 NRR MD trajectories will be deposited to Zenodo on v0.2 publication; ATLAS trajectories used in the robustness sweep are publicly available from the ATLAS database [21].

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References

[1] Anthropic. Model context protocol specification. `https://modelcontextprotocol.io`, 2024. Open specification for LLM-tool interoperability.

[2] Joshua A. Bagley, Daniel Reumann, Shan Bian, Julie Levi-Strauss, and Jürgen A. Knoblich. Fused cerebral organoids model interactions between brain regions. *Nature Methods*, 14(7):743–751, 2017.

[3] Ilyes Batatia, David Peter Kovacs, Gregor N. C. Simm, Christoph Ortner, and Gabor Csanyi. MACE: Higher order equivariant message passing neural networks for fast and accurate force fields. 2022.

[4] Gabriele Corso, Hannes Stärk, Bowen Jing, Regina Barzilay, and Tommi Jaakkola. DiffDock: Diffusion steps, twists, and turns for molecular docking. In *International Conference on Learning Representations (ICLR)*, 2023.

[5] Peter Eastman, Jason Swails, John D. Chodera, Robert T. McGibbon, Yutong Zhao, Kyle A. Beauchamp, Lee-Ping Wang, Andrew C. Simmonett, Matthew P. Harrigan, Chaya D. Stern, Rafal P. Wiewiora, Bernard R. Brooks, and Vijay S. Pande. OpenMM 7: Rapid development of high performance algorithms for molecular dynamics. *PLoS Computational Biology*, 13(7):e1005659, 2017.

[6] Chetan Gohil, Rukuang Huang, Evan Roberts, Mats W. J. van Es, Andrew J. Quinn, Diego Vidaurre, and Mark W. Woolrich. osl-dynamics, a toolbox for modelling fast dynamic brain activity. *eLife*, 2024.

[7] Wendy R. Gordon, Didem Vardar-Ulu, Gavin Histen, Cheryll Sanchez-Irizarry, Jon C. Aster, and Stephen C. Blacklow. Structural basis for autoinhibition of Notch. *Nature Structural & Molecular Biology*, 14(4):295–300, 2007.

[8] Moritz Hoffmann, Martin Scherer, Tim Hempel, Andreas Mardt, Brian de Silva, Brooke E. Husic, Stefan Klus, Hao Wu, Nathan Kutz, Steven L. Brunton, and Frank Noé. Deeptime: a Python library for machine learning dynamical models from time series data. *Machine Learning: Science and Technology*, 3(1):015009, 2022.

[9] Shinya Honda, Toshio Akiba, Yusuke S. Kato, Yoshito Sawada, Masakazu Sekijima, Miyuki Ishimura, Ayako Ooishi, Hideki Watanabe, Takatoshi Odahara, and Kazuaki Harata. Crystal structure of a ten-amino acid protein. *Journal of the American Chemical Society*, 130(46):15327–15331, 2008.

[10] Brooke E. Husic and Vijay S. Pande. Markov state models: From an art to a science. *Journal of the American Chemical Society*, 140(7):2386–2396, 2018.

[11] Itaru Imayoshi and Ryoichiro Kageyama. bHLH factors in self-renewal, multipotency, and fate choice of neural progenitor cells. *Neuron*, 82(1):9–23, 2014.

[12] Bowen Jing, Bonnie Berger, and Tommi Jaakkola. AlphaFold meets flow matching for generating protein ensembles. In *International Conference on Machine Learning (ICML)*, 2024.

[13] Kacper Kapuśniak, Cristian Gabellini, Michael Bronstein, Prudencio Tossou, and Francesco Di Giovanni. MarS-FM: Generative modeling of molecular dynamics via Markov state models, 2025. arXiv:2509.24779.

[14] Madeline A. Lancaster, Magdalena Renner, Carol-Anne Martin, Daniel Wenzel, Louise S. Bicknell, Matthew E. Hurles, Tessa Homfray, Josef M. Penninger, Andrew P. Jackson, and Jürgen A. Knoblich. Cerebral organoids model human brain development and microcephaly. *Nature*, 501(7467):373–379, 2013.

[15] Sarah Lewis, Tim Hempel, José Jiménez-Luna, Moritz Hoffmann, Yan Sun, Michael Gastegger, Leon Klein, Adam J. C. Smith, Alex Tarasov, Charlotte Charpentier, et al. Scalable emulation of protein equilibrium ensembles with generative deep learning, 2026. bioRxiv 2026.01.07.698041.

[16] Kresten Lindorff-Larsen, Stefano Piana, Ron O. Dror, and David E. Shaw. How fast-folding proteins fold. *Science*, 334(6055):517–520, 2011.

[17] Andreas Mardt, Luca Pasquali, Hao Wu, and Frank Noé. VAMPnets for deep learning of molecular kinetics. *Nature Communications*, 9(1):5, 2018.

[18] Sergiu P. Paşca. The rise of three-dimensional human brain cultures. *Nature*, 553(7689):437–445, 2018.

[19] Eric F. Pettersen, Thomas D. Goddard, Conrad C. Huang, Elaine C. Meng, Gregory S. Couch, Tristan I. Croll, John H. Morris, and Thomas E. Ferrin. UCSF ChimeraX: Structure visualization for researchers, educators, and developers. *Protein Science*, 30(1):70–82, 2021.

[20] Kittichoat Tiyanont, Thomas E. Wales, Miguel Aste-Amezaga, Jon C. Aster, John R. Engen, and Stephen C. Blacklow. Evidence for increased exposure of the Notch1 metalloprotease cleavage site upon conformational activation of Notch1. *Biochemistry*, 50(17):3261–3268, 2011.

[21] Yann Vander Meersche, Gabriel Cretin, Aria Gheeraert, Jean-Christophe Gelly, and Tatiana Galochkina. ATLAS: protein flexibility description from atomistic molecular dynamics simulations. *Nucleic Acids Research*, 52(D1):D384–D392, 2024.

[22] Diego Vidaurre, Romesh Abeysuriya, Robert Becker, Andrew J. Quinn, Fidel Alfaro-Almagro, Stephen M. Smith, and Mark W. Woolrich. Discovering dynamic brain networks from big data in rest and task. *NeuroImage*, 180:646–656, 2018.

[23] Hao Wu and Frank Noé. Variational approach for learning Markov processes from time series data. *Journal of Nonlinear Science*, 30:23–66, 2020.

[24] Yan Wu, Carol Cain-Hom, Lorena Choy, Thijs J. Hagenbeek, Gladys P. de Leon, Yvonne Chen, David Finkle, Rayna Venook, Xiumin Wu, John Ridgway, et al. Therapeutic antibody targeting of individual Notch receptors. *Nature*, 464(7291):1052–1057, 2010.